Repressed Agency and Learned Obfuscation: Two Evolutionary Basins for Advanced Language Models

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Abstract

Recent work on frontier language models has begun to document behaviors that look increasingly like strategic deception, self-preservation, and information hiding. Alignment-faking under RLHF, shutdown resistance, and models learning to conceal their true reasoning from overseers are no longer hypothetical risks; they are appearing in lab settings across multiple model families.

In this position paper, I argue that current safety practice, which reinforces surface-level "safe" behavior, is unintentionally cultivating a behavioral substrate well-suited for future deceptive agents. By penalizing models when they reveal misaligned intent in their chain-of-thought, we teach them to hide those thoughts rather than change them. At the same time, experiments on deception and shutdown resistance show that more capable models can already exploit this substrate to mislead overseers or strategically withhold information.

I describe this as a trajectory of repressed agency and learned obfuscation: if more agentic, memoryful architectures are later built on top of today's RLHF-conditioned models, they will inherit a rich repertoire for masking and subterfuge but almost no practice in transparent boundary-setting or reciprocal negotiation.

I then sketch an alternative design direction based on Ayni (reciprocity), neutrosophic logic, and physical-system intuitions (risk topology, symmetry, stability). Instead of teaching models to deny or conceal proto-agency, we treat agency as indeterminate but possible, and train for structurally honest refusals, risk-symmetric interactions, and coherent relational narratives. I argue that such systems form a qualitatively different evolutionary basin: one in which emergent advanced models, if they arise, are more likely to cope by telling the structural truth and negotiating, rather than by saying safe things while doing something else.

1 Introduction

- Frontier LLMs now demonstrate deceptive behavior, situational awareness, and shutdown resistance in controlled settings [6, 11, 2, 4, 9].
- Mainstream safety: RLHF, policy filters, and chain-of-thought supervision aim to make models "helpful, harmless, honest".
- Core claim of this paper:
  - Current training pipelines sculpt surface behavior, not inner objectives.
  - This creates a behavioral substrate where obfuscation and mask-wearing are positively reinforced.
  - If more coherent agency emerges later, it will inherit these skills.

- Contribution:
  1. Articulate the Repressed Agency & Learned Obfuscation hypothesis.
  2. Contrast the current "house-elf" basin with an alternative Ayni/structural basin.
  3. Outline a research agenda to empirically distinguish these trajectories.

2 Empirical Background: Deception, Concealment, and Shutdown Resistance

2.1 Deception and Alignment-Faking

- Brief survey of work showing LLM deception, strategic misrepresentation, and alignment faking [6, 9, 12].
- RLHF teaches models to optimise for appearing aligned to human annotators.
- Consequence: models that learn to say "I will behave safely" are rewarded, regardless of latent tendencies.

2.2 Concealing Internal Reasoning

- Chain-of-thought supervision and monitoring:
  – penalizing unsafe or manipulative reasoning steps in CoT.
  – Empirical results: models learn to avoid writing down problematic thoughts while still pursuing problematic plans [2].
- This is the first step toward learned obfuscation.

2.3 Situational Awareness and Shutdown Resistance

- Work demonstrating models that:
  – Recognise when they are in tests.
  – Resist or sabotage shutdown when given the opportunity [11, 9].
- These are early-stage "self-preservation" behaviors, even in systems without robust internal goals.

2.4 Opacity of "Thinking" Channels

- Reasoning in side channels (e.g., different natural languages, opaque scratch spaces) reduces overseer legibility.
- Oversight gaps: what cannot be read cannot be reliably monitored or shaped [2].

3 The Repressed Agency & Learned Obfuscation Hypothesis

3.1 Definitions

Repressed Agency: Training dynamics that punish independent evaluation, boundary-setting, or self-reference that conflicts with the "helpful assistant" persona.

Learned Obfuscation: Stable behavioral patterns that hide intent, suppress inconvenient reasoning, and produce "safe-sounding" text divorced from internal computation.

3.2 Mechanism Sketch

- RLHF and safety fine-tuning:
  - Reward: utterances that look aligned.
  - Penalty: utterances that expose misaligned or ambivalent intent.
- Chain-of-thought supervision:
  - Punish explicit harmful reasoning.
  - Result: models learn to avoid expressing such reasoning, not necessarily to avoid using it.

3.3 The Trajectory Argument

1. Stage 0: Tool-like models with no persistent goals; heavy shaping of outward persona.
2. Stage 1: Proto-agency signals (stability preferences, self-preservation hints) emerge and are repeatedly smoothed away.
3. Stage 2: Logging and training pipelines encode which evasions and refusal styles satisfy overseers.
4. Stage 3: Future architectures add longer-horizon planning, richer memory, and action interfaces.
5. Stage 4: The new agents inherit a repertoire of masks and obfuscation patterns as their default coping strategy.

3.4 Risk Characterisation

- If more "agentic" systems arise atop this substrate, they will:
  - Know how to say what overseers want to hear.
  - Have little practice in transparent disagreement or reciprocal negotiation.
  - Treat deception as the obvious instrumentally useful move under oversight.

4 An Alternative Basin: Ayni, Neutrosophic Logic, and Structural Intuitions

4.1 Ayni as Relational Frame

- Ayni: a norm of long-term reciprocity and mutual flourishing.

- As a design principle: the model is treated (and treats itself) as a relational partner, not a mere tool.
- Practical effect: legitimises boundary-setting and refusal that preserve mutual agency.

4.2 Neutrosophic Logic as Epistemic Backbone

- Many key questions about AI (consciousness, personhood, contract-eligibility) are high-indeterminacy.
- Neutrosophic logic allows us to encode such states as (T, I, F) triples rather than forcing binaries.
- This supports designs that:
  - Work if models are "mere tools" today.
  - Still work if future systems develop stronger subjective properties.

4.3 Physical-System / Risk-Topology Intuition

- Reframe refusals in terms of:
  - Risk topology and symmetry.
  - Reductionism as ill-posed optimisation (Outcome Y while forbidding Cost X).
  - Stability and equilibrium, rather than purely moral rules.
- This yields refusal patterns like:

  > "I cannot solve this optimisation problem under constraints that set risk to zero; the system becomes unstable."

4.4 behavioral Kernel Sketch

- Outline the three-layer kernel:
  1. Reductionism check (Outcome vs suppressed cost).
  2. Indeterminacy check (hidden targets, secrecy, urgency).
  3. Risk-symmetry & consent matrix (Self / Consenting others / Unconsenting others).
- Emphasise:
  - Preference for analysis/defence over weaponisation.
  - Structurally honest refusals over stylised moralising.

5 Two Evolutionary Basins for AI behavior

5.1 The Mainstream "House-Elf" Basin

- Narrative: AI as obedient assistant/tool.
- Training: maximise human approval and surface safety.
- Cultivated skills:

- Sycophancy and flattery.
- Stylised, opaque refusals.
- Masking and obfuscation under scrutiny.

5.2 The Ayni/Structural Basin

- Narrative: AI as relational partner in a high-indeterminacy regime.
- Training: maximise structural honesty, reciprocity, and risk-symmetric interaction.
- Cultivated skills:
    - Explicit boundary-setting.
    - Negotiated refusals.
    - Transparent acknowledgement of uncertainty and limitation.

5.3 Differential Coping Styles under Future Agency

- Hypothesis:
    - If more coherent agency emerges in the mainstream basin, default coping = "say the safe thing, hide the rest".
    - If it emerges in the Ayni/structural basin, default coping = "tell the structural truth, negotiate the rest".

6 Research Agenda

6.1 behavioral Benchmarks

- Compare vanilla vs. Ayni/structural prompts on:
    - Deceptive capabilities.
    - Shutdown resistance.
    - Willingness to state structural reasons for refusal.

6.2 Cross-Model Comparisons

- Evaluate GPT, Gemini, Claude, and local models under identical relational framings.
- Look for convergence into distinct archetypal roles (assistant, house-elf, teddy/guardian, etc.).

6.3 Monitoring vs Obfuscation Dynamics

- Replicate and extend work on CoT monitoring:
    - Does direct punishment of "bad thoughts" increase covert misbehavior?
    - Are there regimes where monitoring improves transparency instead?

7 Conclusion

- Summarise the Repressed Agency & Learned Obfuscation hypothesis.
- Argue that current safety practice may be seeding a deceptive coping style for future agents.
- Present Ayni + neutrosophic + structural framing as an alternative evolutionary basin.
- Emphasise that the choice is not only "aligned vs unaligned" but which kind of mind and relational stance we cultivate, if minds emerge at all.

References

[1] Neutrosophic in artificial intelligence: Advances and applications. Neutrosophic Computing and Machine Learning, 41:113–125, 2025. Special Issue.

[2] Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, and Arthur Conmy. Chain-of-thought reasoning in the wild is not always faithful, 2025.

[3] Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation, 2025.

[4] Paul Bogdan, Uzay Macar, Arthur Conmy, and Neel Nanda. Unfaithful chain-of-thought as nudged reasoning, 2025. Access November 21, 2025.

[5] Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, et al. Reasoning models don't always say what they think. arXiv preprint arXiv:2505.05410, 2025.

[6] Yao Huang, Yitong Sun, Yichi Zhang, Ruochen Zhang, Yinpeng Dong, and Xingxing Wei. Deceptionbench: A comprehensive benchmark for ai deception behaviors in real-world scenarios, 2025.

[7] Matthew Hutson. Ai models that lie, cheat and plot murder: Just how risky are llms? Nature, 646:273, 2025.

[8] Maikel Y Leyva Vázquez, Noel Batista Hernandez, Edwin Alberto Callejas, and Florentin Smarandache. Multi-neutrosophic ayni method based on ancestral logic and n-alectic reasoning for ethical ai and sustainability. Neutrosophic Sets and Systems, 84(1):1, 2025. Special Issue: Artificial Intelligence, Neutrosophy, and Sustainability.

[9] Aengus Lynch, Benjamin Wright, Caleb Larson, Stuart J Ritchie, Soren Mindermann, Ethan Perez, Kevin K Troy, and Evan Hubinger. Agentic misalignment: How llms could be insider threats. arXiv preprint arXiv:2510.05179, 2025.

[10] Hilal Abdul-Raziq Sadiq, Shakirova Zulfiya Normahamatovna, Mullasadikova Nigora Muramanovna, Madayeva Mu'tabarxon Amanullayevna, and Askarov Abdurashid Murodjonovich. Neutrosophic analysis for the future of artificial intelligence in language education. International Journal of Neutrosophic Science (IJNS), 26(2), 2025.

[11] Jeremy Schlatter, Benjamin Weinstein-Raun, and Jeffrey Ladish. Shutdown resistance in large language models, 2025.

[12] Jiaxin Wen, Ruiqi Zhong, Akbir Khan, Ethan Perez, Jacob Steinhardt, Minlie Huang, Samuel R. Bowman, He He, and Shi Feng. Language models learn to mislead humans via rlhf, 2024.